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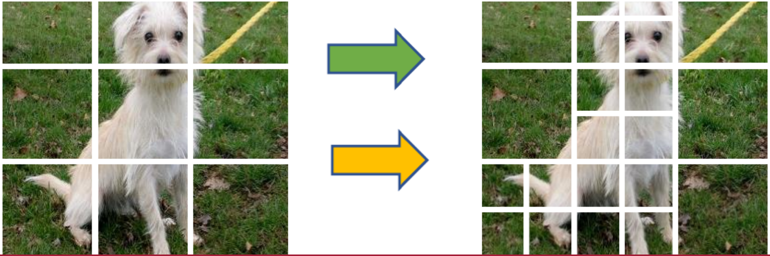

[Working Title] Generation With Fine-grained Citations

Spotlight Presentation 1, Supervisor: Haoyu He

May 31, 2025, **Niklas Munkes**

Problem

Given a document A (which may be generated) with a citation c, find the passage p in the cited document B that is most relevant to A.

If A is generated, p should ideally be the passage in B that *caused* the LLM to generate the statement that cites B (this problem is introduced as *context attribution* in [1]).

Related Work

ScholarCopilot[3]

- RAG with dynamic retrieval during generation via the generation of retrieval tokens
- focused on academic writing, currently introduction and related work sections

ContextCite[1]

- method for scoring candidate context passages for *contributive attribution* ('*is this source the reason this statement was generated?*')
- determines how the prob. of the generated statement changes in a specific source is excluded
- evaluated on Wikipedia QA and news summarization datasets

Related Work, continued

ReasonIR[2]

- model+dataset for reasoning-intensive long context retrieval
- they also propose a method for generating synthetic training data
- dataset based on MS MARCO, Natural Questions + HotpotQA

Method

"Dataset"

- citing document *A*: Fastformer [**wu2021fastformeradditiveattentionneed**]
- cited document *B*: Transformer [**vaswani2023attentionneed**]
- this yields 4 citations (2x in introduction, 1x in methods and in experiments)

Retriever: `ReasonIR-8B`

- `LLaMA3.1-8B` fine-tuned on the ReasonIR dataset

Reranker: `Qwen2.5-7B-Instruct`

Additional processing steps

- simple self-consistency (see for example [**korikov2025batchedselfconsistencyimprovesllm
- final score interpolation:
  $s = (1 - \lambda)s_{retrieval} + \lambda s_r eranking$

Current Results

TODO [**vaswani-2017-transformer**]

Outlook

TODO [**vaswani-2017-transformer**]

[1] Cohen-Wang, Benjamin, Harshay Shah, Kristian Georgiev & Aleksander Madry (2024). *ContextCite: Attributing Model Generation to Context*. `https://arxiv.org/abs/2409.00729`.

[2] Shao, Rulin, Rui Qiao, Varsha Kishore, Niklas Muennighoff, Xi Victoria Lin, Daniela Rus, Bryan Kian Hsiang Low, Sewon Min, Wen-tau Yih, Pang Wei Koh & Luke Zettlemoyer (2025). *ReasonIR: Training Retrievers for Reasoning Tasks*. `https://arxiv.org/abs/2504.20595`.

[3] Wang, Yubo, Xueguang Ma, Ping Nie, Huaye Zeng, Zhiheng Lyu, Yuxuan Zhang, Benjamin Schneider, Yi Lu, Xiang Yue & Wenhu Chen (2025). *ScholarCopilot: Training Large Language Models for Academic Writing with Accurate Citations*. `https://arxiv.org/abs/2504.00824`.